

LESSON 1.2

ADDING AND SUBTRACTING FRACTIONS



LESSON INTRODUCTION

MA.5.AR.1.2 Solve real-world problems involving the addition, subtraction, or multiplication of fractions, including mixed numbers and fractions greater than 1.

LEARNING TARGETS

- I can solve real-world problems involving adding and subtracting fractions.

BENCHMARK COHERENCE

WORKING TOWARD

MA.6.NSO.2.3 Solve multistep real-world problems involving any of the four operations with positive multidigit decimals or positive fractions, including mixed numbers.

ESSENTIAL QUESTIONS

- How can you use models to add and subtract fractions that have different denominators?
- How can multiples be used to rewrite fractions with a common denominator?
- What are the steps for adding or subtracting fractions with unlike denominators?
- Why is it necessary to rewrite fractions with common denominators before adding or subtracting?

LESSON NARRATIVE

In this lesson, students review how to add and subtract fractions and mixed numbers that have unlike denominators. Students activate background knowledge using multiples to rewrite fractions with unlike denominators as fractions with like denominators, then practice adding fractions in a real-world scenario. The lesson wraps up with students practicing rewriting fractions and mixed numbers to have common denominators to solve both real-world and mathematical problems.

COMPONENT SUMMARY

1.2.1 WARM-UP (~5 MIN.)

Students write a true statement with fractions equal to one

1.2.2 EXPLORATION (10-15 MIN.)

Exploration on adding fractions in context of framing an art piece

1.2.3 GUIDED INSTRUCTION (15-20 MIN.)

Students continue using context of framing artwork to solve for width and height using fraction addition

1.2.4 GUIDED PRACTICE (10-15 MIN.)

Practice adding and subtracting with unlike denominators and mixed fractions

1.2.5 WRAP-UP (~5 MIN.)

Assessment on determining common denominators and solving addition and subtraction fraction problems

RESOURCES

GLOSSARY

Key Terms:

- Multiplicative Identity Property of One

Additional Terms:

- Equivalent fraction
- Multiple, Factor
- Common denominator

MATERIALS

- (Optional) Fraction manipulatives
- (Optional) Paper/artwork for modeling framed art

ADDITIONAL PREPARATION

- If using math fraction manipulatives, drawing materials for models, or other methods for visualizing fraction addition and subtraction, gather enough materials for group of students prior to lesson.
- (Optional) Gather cardboard and “artwork” pieces that students can use to measure, label, and cut out to visualize the context in 1.2.2 Exploration

TEACHER TIPS

COMMON MISCONCEPTIONS

- Students may have difficulty converting mixed numbers into fractions greater than one.
- Students may have difficulty finding common denominators between the different addends.
- Students may not know or remember that fractions with common denominators are equivalent fractions.

THINGS TO CONSIDER

- Encourage students to draw a model or use fraction manipulatives to help support converting mixed numbers into improper fractions greater than one.
- Position students to relate common denominators to factors and multiples of numbers, which may help students rewrite fractions with lowest common denominators and consequently simplify arithmetic work required.

LITERACY AND LANGUAGE ROUTINES

Compare and Connect

This lesson contains rich vocabulary related to fractions and provides opportunities to represent and approach solving addition and subtraction fraction problems in a variety of methods. Consider having students use and phrase thinking using correct vocabulary to build conceptual understanding and to compare approaches to finding common denominators to foster awareness of procedural approaches.

MATHEMATICAL THINKING AND REASONING (MTR) STANDARD

MA.K12.MTR.7.1 Real-World Contexts

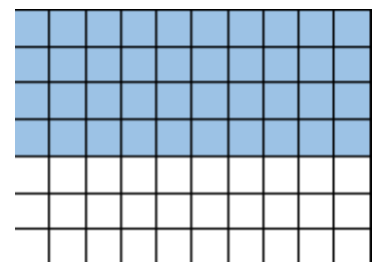
In 1.2.3 Guided Instruction, students interact with addition and subtraction problems in the real world that use fractions. The problems require students to connect mathematical concepts to everyday experiences and to use models to understand, represent, and solve problems. Consider encouraging student understanding by providing pictures and cardboard frames.

DIFFERENTIATE INSTRUCTION

Consider using or encouraging students to create visual models throughout this lesson to represent the addition or subtraction with fractions or to show how fractions are equivalent with unlike or like denominators. Using models or visuals to support vocabulary will be important to help all learners understand and connect the conceptual and procedural learning targets.

In 1.2.1 Warm-Up and 1.2.3 Guided Instruction, consider using kinesthetic approaches, including modeling with fraction manipulatives, to help students see fraction equivalence and methods for rewriting fractions with common denominators.

$$4\frac{4}{7}$$



1.2.1 WARM-UP: TRUE STATEMENT

Component Overview: Prompt students to read and attempt the question in the Warm-Up independently using each digit no more than once.

ENRICHMENT

Some students may struggle in completing this problem because the two fractions they are asked to create will not have the same denominator. Since this lesson rests on students being able to find equivalent fractions, consider prompting students to use unit fractions first, then make one into an equivalent fraction using a different denominator.

To support student thinking about fractions whose sum is one, scaffold using questions like:

If the fractions need to add up to one, what do you know about the values?

- How do we add fractions with like denominators? With unlike denominators?
- How can you use equivalent fractions to rewrite values using different digits?
- How can you draw a model to confirm your fractions add up to one?



1.2.1 Warm-Up: True Statement

1. Place a digit in each blue box to make the statement true. Use only digits 1 to 9, at most one time each.

$$\frac{\boxed{}}{\boxed{}} + \frac{\boxed{}}{\boxed{}} = 1$$

Answers will vary. Correct answers include:

$$\frac{6}{9} + \frac{1}{3} = 1$$

$$\frac{4}{4} + \frac{1}{1} = 1$$

$$\frac{7}{8} + \frac{1}{2} = 1$$

$$\frac{6}{8} + \frac{1}{4} = 1$$



1.2.2 Exploration: Framing a Piece of Art

Claudia bought a piece of art at the Winter Park Sidewalk Art Festival. She wanted to build her own frame for the piece of art. The piece of art has a height of $1\frac{1}{3}$ feet (ft) and a width of $\frac{3}{4}$ ft.

1. Claudia's piece of art is shown. Sketch a frame that Claudia could use. Label the width of the frame.
Answers will vary.



The sketches of the frames of three classmates are shown.

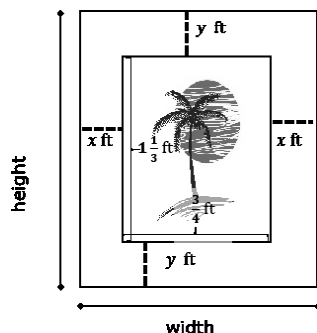
Eduardo	Jahmelia	Joshua

2. Which of these frames, when combined with the art piece, will produce a height that is 4 ft taller than the original piece of art?
Jahmelia and Joshua

3. Whose frame, when combined with the art piece, will produce a width that is 1 ft longer than the original art piece?
Joshua

The frames of Eduardo, Jahmelia, and Joshua show that the width of the frame can vary. Joshua's frame shows that the frame does not have to be the same width on all four sides.

4. Claudia wants to determine the amount of wood she needs to frame the piece of art. Since she is not sure of the widths of the frame, she labels the dimensions, as shown.



- a. Work with your partner to write an expression that represents the width.
Answers will vary. $\frac{3}{4} + x + x$. $\frac{3}{4} + 2x$
- b. Work with your partner to write an expression that represents the height.
Answers will vary. $1\frac{1}{3} + y + y$. $1\frac{1}{3} + 2y$

1.2.2 EXPLORATION: FRAMING A PIECE OF ART

Component Overview: In this activity, students compare three different fraction dimensions of frames for a given piece of art, considering the dimensions with and without the frame. Place students in partnerships as they explore this real-world scenario, create expressions to represent the framed artwork, and solve for the height and width of the art piece with a frame.

DIFFERENTIATED INSTRUCTION

Students may not immediately understand that a frame surrounds the art, which is an important entry point for this real-world scenario. Consider providing a tangible example of a piece of art framed to help visually contextualize the scenario. Also consider having students create a real-world example of Claudia's art piece by measuring and labeling a sheet of paper as another visual entry point.

ENRICHMENT

To help students know to add values and recognize how to use the given dimensions in upcoming questions, ask students:

- What is the difference between tall and wide?
- How do these descriptors relate to the dimensions of the frame?
- How do these descriptors help you know which operation to use?

ENRICHMENT

Notice the use of variables to represent an unknown value. For some students without prior exposure to variables in mathematics, writing an expression to represent the width may be challenging. Consider asking scaffolded questions to support students' exploration, like:

- Even though we do not know the value, what operation would we need to use to know the total distance of the width?
- How can you use that operation to write an expression with x ?

TEACHER MOVES

Compare and Connect

Before students share their thinking with a partner, prompt students to first write an expression representing the width of the frame independently. Then as students share, have students discuss whether their expressions represent the same or different scenarios. This approach may help students recognize that there is more than one way to represent the width in this scenario.

1.2.3 GUIDED INSTRUCTION: ADDING AND SUBTRACTING FRACTIONS WITH UNLIKE DENOMINATORS

Component Overview: In this activity, students continue with the same context of the art piece, relating algorithms to add and subtract fractions and mixed numbers to find common denominators to be able to find the height and width of the entire art piece. Students list common denominators to find the smallest common denominator, then use that value to make equivalent fractions to find the total width and height. Work as a class through this section to address any misconceptions and ensure students are finding and using correct common denominators.

ENRICHMENT

Students may not recognize the difference between factors and multiples and why multiples are used to write/identify common denominators. Position students to recognize the difference by asking:

- What is the difference between a factor and a multiple?
- Can you provide an example of a factor and a multiple for the value 10?

TEACHER MOVES

Consider using visual or kinesthetic approaches for question 1, including drawing a model or using fraction manipulatives to strengthen understanding of writing common denominators. Facilitate a discussion based on these models to help students relate common denominators to multiples of numbers, asking students to explain why the smallest multiple is better to use rather than simply multiplying the denominators to find a common denominator.

DIFFERENTIATED INSTRUCTION

Consider engaging in a Think Aloud as you explain or reason about why the circled fractions are equivalent fractions to one whole. This auditory approach can be supplemented with visual models or fraction manipulatives to represent this equivalency. Then verbalize how to connect these representations to the fraction forms and algorithms as needed.

TEACHER MOVES

Compare and Connect

Have students share their ideas for questions 3 and 4 as a whole class, focusing on comparing the explanations and reasoning related to finding common denominators. Encourage students to use accurate vocabulary words when describing parts or actions related to fraction operations.



1.2.3 Guided Instruction: Adding and Subtracting Fractions with Unlike Denominators

One way to add or subtract fractions, is to first find **equivalent** fractions that have **common**, or **the same** denominators.

To find a common denominator for fractions write the **factors** or **multiples** of the numbers in the denominators.

1. The table shows the number in each denominator of the fractions that represents the width and height of Claudia's art piece.

- a. List at least 10 multiples of each.

3	3, 6, 9, 12, 15, 18, 21, 24, 27, 30, 33
4	4, 8, 12, 16, 20, 24, 28, 32, 36, 40, 44

- b. Circle the smallest multiple that both denominators have in common.
Answer shown in table.

2. Working with only the fractional portion of each number, multiply both the numerator and the denominator by the factor that will result in the circled denominator.

$$\frac{1}{3} \times \frac{4}{4} = \frac{4}{12}$$

$$\frac{3}{4} \times \frac{3}{3} = \frac{9}{12}$$

The circled fraction is equivalent to 1. Multiplying by 1 means that the value of the original fraction remains the same.



Multiplicative Identity Property of One

$$a \times 1 = a$$

$$1 \times a = a$$

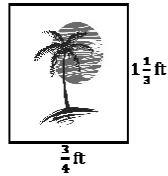
3. Rewrite $1\frac{1}{3}$ using the equivalent fraction found for $\frac{1}{3}$ by completing the fractions.

Because $1\frac{1}{3} = \frac{3}{3} + \frac{1}{3}$ then $1\frac{1}{3}$ can be rewritten using the equivalent fractions $\frac{3}{3} + \frac{1}{3} = \frac{12}{12} + \frac{4}{12}$.

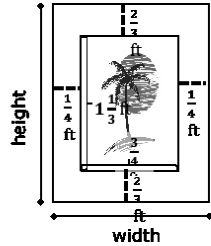
4. Claudia's art piece is shown.

- a. What is the perimeter, in feet (ft), of Claudia's art piece?

$$4\frac{1}{6} \text{ ft}$$



Claudia decides to build a frame with the following dimensions.



- b. Write the numerical expression that represents the total width of the frame and art piece.

$$\frac{3}{4} + \frac{1}{4} + \frac{1}{4}$$

- c. Find the total width, in ft.

$$1\frac{1}{4} \text{ ft}$$

- d. Write the numerical expression that represents the total height of the frame and art piece.

$$1\frac{1}{3} + \frac{2}{3} + \frac{2}{3}$$

- e. Find the total height, in ft.

$$2\frac{2}{3} \text{ ft}$$

1.2.3 GUIDED INSTRUCTION: ADDING AND SUBTRACTING FRACTIONS WITH UNLIKE DENOMINATORS (CONTINUED)

ENRICHMENT

Position students to better understand why and how to rewrite question 4 with common denominators by asking:

- What multiples would help you find a common denominator?
- How are you using the Multiplicative Identity Property of One as you rewrite the fractions with common denominators?

DIFFERENTIATED INSTRUCTION

Question 4a requires students to make a mixed number into an improper fraction before adding. Consider reviewing the conceptual reasoning and procedural process for rewriting mixed numbers as improper fractions by having students draw a fraction model or counting unit fractions with a denominator of 12.

TEACHER MOVES

To foster metacognitive thinking, ask students to explain if their answers are reasonable for questions 4c and 4e given the context.

1.2.4 GUIDED PRACTICE: ADDING AND SUBTRACTING FRACTIONS WITH UNLIKE DENOMINATORS

Component Overview: Students take the previous practice within real-world contexts to now focus on the process for adding and subtracting both mixed numbers and fractions by first writing the fractions with common denominators.

QUESTIONING

Consider having students work independently to attempt the four problems first, then as a class review the correct answers and ask individual students to share their arithmetic and thinking with the group.

TEACHER MOVES

As students share their answers and thinking with the group, encourage them to verbalize the steps they took to find the sums or differences while using correct math vocabulary related to fractions. Consider fostering this math talk around both procedural and conceptual skills by using sentence stems like:

- “To find common denominators, first I...”
- “I know the fractions are equivalent because...”
- “I used multiples to find common denominators by...”
- “I found the answer by first...then...next...”
- “I know my answer is reasonable because...”



1.2.4 Guided Practice: Adding and Subtracting Fractions with Unlike Denominators

Rewrite the following expressions as an expression with common denominators. Then, add or subtract the values.

$$1. \quad 4\frac{1}{3} - \frac{3}{5} =$$

$$\frac{65}{15} - \frac{9}{15} = \frac{56}{15} \text{ or } 3\frac{11}{15}$$

$$2. \quad 1\frac{7}{10} + 6\frac{3}{4} =$$

$$\frac{34}{20} + \frac{135}{20} = \frac{169}{20} \text{ or } 8\frac{9}{20}$$

$$3. \quad \frac{3}{5} + 1\frac{9}{10} =$$

$$\frac{6}{10} + \frac{19}{10} = \frac{25}{10} = \frac{5}{2} \text{ or } 2\frac{1}{2}$$

$$4. \quad 2\frac{1}{2} - 1\frac{1}{4} =$$

$$\frac{26}{12} - \frac{15}{12} = \frac{11}{12}$$



1.2.5 Wrap-Up: Ticket Out the Door

1. Use the table below to answer the questions.

Complete the table. First, find the common denominator for **Fraction A** and **Fraction B**. Then, rewrite **Fraction A** and **Fraction B** using the common denominator.

Fraction A	Fraction B	Common Denominator	Equivalent Fractions with a Common Denominator	
			A:	B:
$\frac{7}{8}$	$\frac{5}{6}$	24	$\frac{21}{24}$	$\frac{20}{24}$
$1\frac{3}{10}$	$3\frac{2}{3}$	30	$\frac{39}{30}$	$\frac{110}{30}$
$5\frac{3}{4}$	$2\frac{1}{3}$	12	$\frac{69}{12}$	$\frac{28}{12}$

2. Find the value of each.

$$\frac{7}{8} + \frac{5}{6} = \frac{41}{24} \text{ or } 1\frac{17}{24}$$

$$5\frac{3}{4} - 2\frac{1}{3} = \frac{41}{12} \text{ or } 3\frac{5}{12}$$

1.2.5 WRAP-UP: TICKET OUT THE DOOR

Component Overview: This Wrap-Up assesses students' abilities to find a common denominator for fractions and mixed numbers by first converting to equivalent fractions, then adding or subtracting. Have students work independently as a check on understanding for this lesson's learning target. Students should not use a calculator to complete this Wrap-Up.

DIFFERENTIATED INSTRUCTION

Consider having students complete all or some of the problems based on individual learning needs and previous familiarity with this lesson's learning targets.

Consider allowing students to create visual methods for showing learning, including using a diagram or model to find the answers in question 2.